

Repo Illustrator

Get to know your repo better



Overview

Project introduction

System flow

Ingestion Pipeline

Insights generation

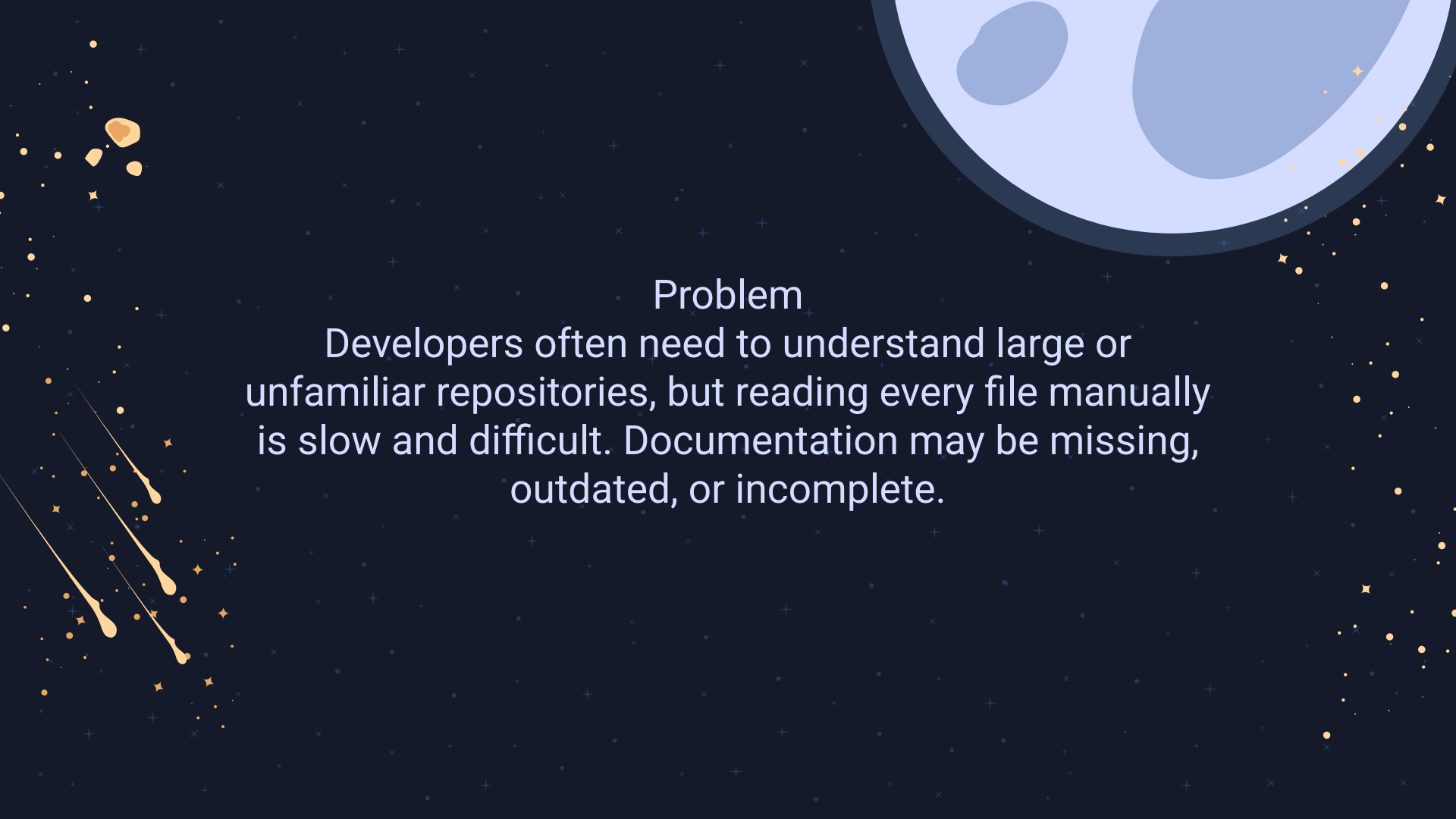
ChatBot functionality

Backend architecture

Frontend architecture

Deployment



The background is a dark blue space scene. In the top right corner, a large, light blue planet with darker blue spots is partially visible. The sky is filled with numerous small, light blue stars and a few larger, yellowish-orange stars. In the bottom left corner, there are three bright yellow-orange comets with long, thin tails pointing towards the top left.

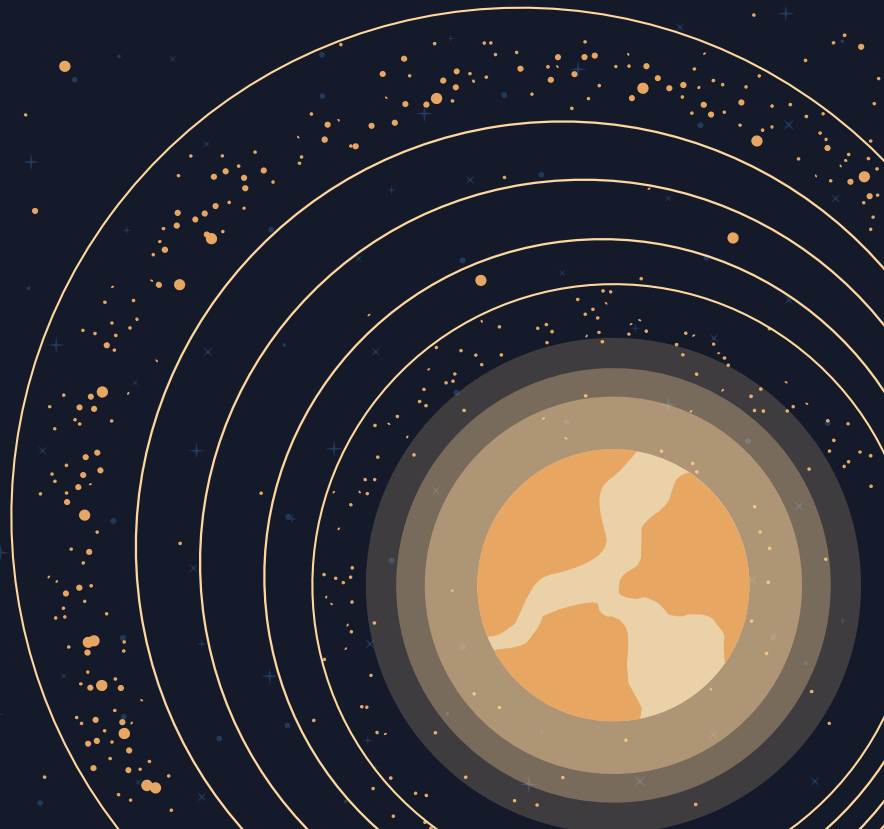
Problem

Developers often need to understand large or unfamiliar repositories, but reading every file manually is slow and difficult. Documentation may be missing, outdated, or incomplete.

Introduction

Repo Illustrator helps users understand unfamiliar codebases faster.

The user provides a GitHub repository URL, and the system analyzes the repository, generates visual insights, creates documentation, and provides a chatbot that can answer questions about the code.



System Flow



User enters a
GitHub repository
URL



Backend starts
repository
ingestion



Repository files
are cloned and
preprocessed



Files are loaded,
cleaned, and split
into chunks

System Flow



Embeddings are generated and stored



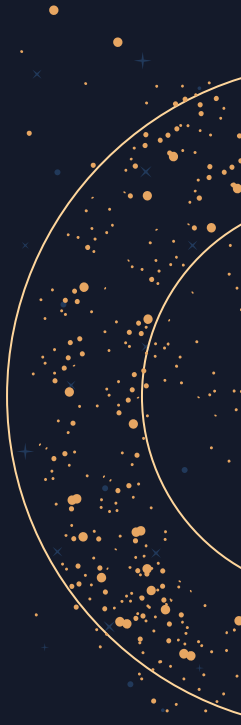
The system analyzes code structure and dependencies



The system generates insights, charts, and documentation

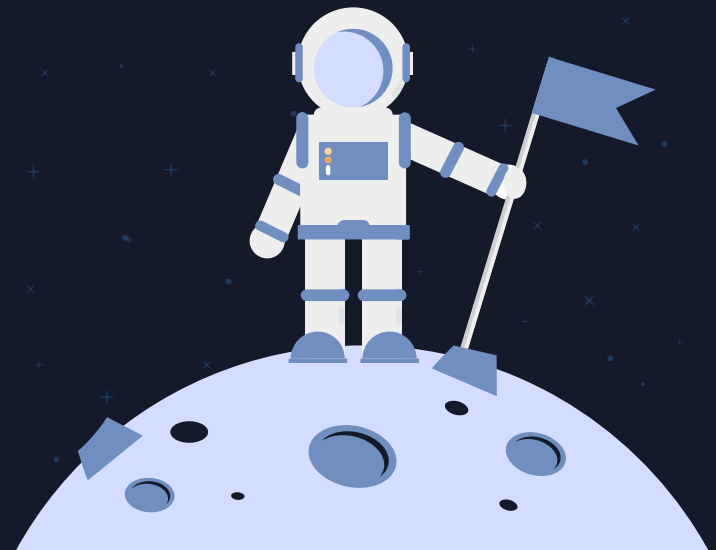


The user interacts with the repo through dashboard and chatbot



Ingestion Pipeline

The ingestion pipeline is responsible for converting a GitHub repository into searchable knowledge.



Ingestion Pipeline

01 Cloning

The repository is cloned from the provided GitHub URL

04 Chunking

Large files are split into smaller chunks using LlamaIndex.

02 Preprocessing

Unneeded files and folders are filtered out to keep only useful project content.

05 Embedding

Each chunk is converted into a numerical embedding.

03 Load Files

The system reads the repository files from the processed folder.

06 Vector Storage

The embeddings are stored in ChromaDB for later retrieval.

Insights

The insights feature generates a high-level understanding of the repository.

It uses an AI agent to inspect important project files such as README files, entry points, dependency files, and source code.



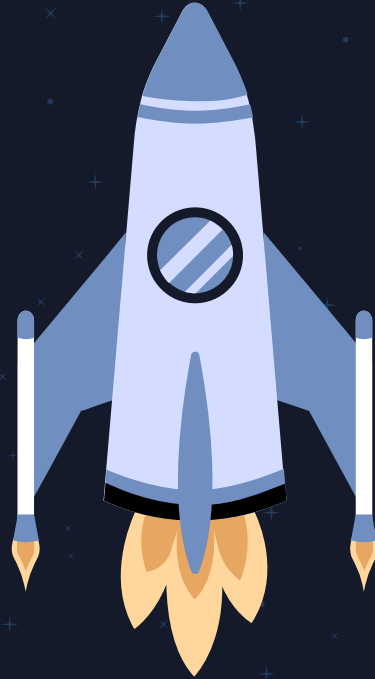
Insights

01 Generates project purpose

02 Detects tech stack

03 Explains architecture

04 Uses GitHub API data for contributor and repository statistics for chart generation



Chatbot

The chatbot allows users to ask natural language questions about the repository. It uses Retrieval-Augmented Generation (RAG), to answer based on the actual codebase instead of relying only on the model's general knowledge. Can answer technical questions about files, architecture, and implementation

Each chat session is saved, and the backend can stream responses for a smoother user experience.





Backend

The backend is built with FastAPI and exposes endpoints for repository ingestion, chat, insights, charts, and documentation.

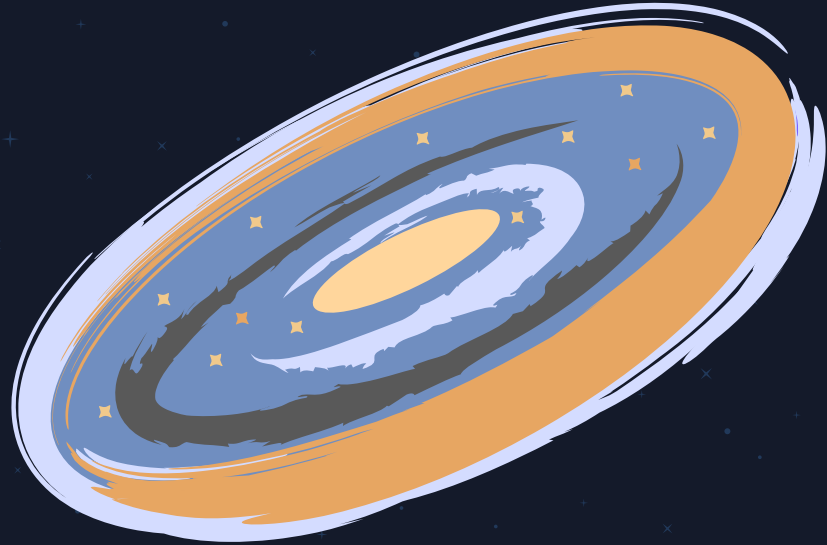
- **Ingestion:** starts repo processing and indexing
- **Chat:** handles normal and streaming AI chat
- **Insight:** generates purpose, tech stack, architecture, and key files
- **Charts:** generates visual analytics data
- **Docs:** generates automatic documentation
- **Database:** stores repo status, chat sessions, cached charts/docs/insights



Frontend

The frontend was built using React with Vite and JavaScript to create a fast, modern, and interactive user interface.

It also includes a “Join our Community” button that connects us with our teammates in the fullstack track.





Repo URL textbox

Chatbot Assistant

Join our community
button

Deployment

The project was containerized using Docker and deployed on Azure to make the application easier to run, manage, and access from different environments.



END